

SUBSTITUTED N-METHYLENE DERIVATIVES OF THIENAMYCIN

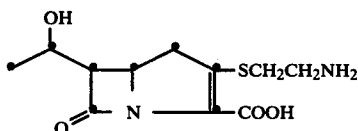
BACKGROUND OF THE INVENTION

This application is a continuation-in-part of U.S. Patent Application Ser. No. 733,654, filed Oct. 18, 1976 now abandoned; which is a continuation-in-part of Application Ser. No. 676,261 filed Apr. 12, 1976 now abandoned, which in turn is a continuation-in-part of Application Ser. No. 634,296, filed Nov. 21, 1975 now abandoned.

This invention relates to certain substituted N-methylene derivatives of the new antibiotic thienamycin. Such compounds and their pharmaceutically acceptable salt, ether, ester, and amide derivatives are useful as antibiotics. This invention also relates to processes for the preparation of such compounds; pharmaceutical compositions comprising such compounds; and methods of treatment comprising administering such compounds and compositions when an antibiotic effect is indicated.

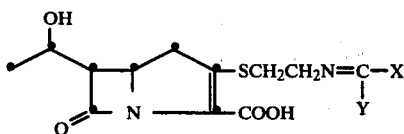
Thienamycin is disclosed and claimed in co-pending, commonly assigned U.S. Patent Application Ser. No. 526,992, filed Nov. 25, 1974 (now U.S. Pat. No. 3,950,357 issued Apr. 13, 1976), which is incorporated herein by reference since thienamycin may be employed as a starting material in the preparation of the compounds of the present invention.

Thienamycin is known to have the following structure:

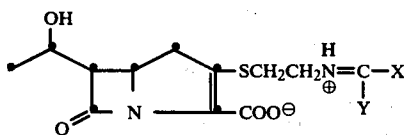


Thienamycin and all of its isomers (in pure form and as mixtures) are also obtainable by the total synthesis disclosed and claimed in co-pending, commonly assigned U.S. Patent Application Ser. No. 833,210 (15 Sept. 1977). This application is incorporated herein by reference to the extent that it makes available all isomers of I as starting materials in the preparation of the compounds of the present invention.

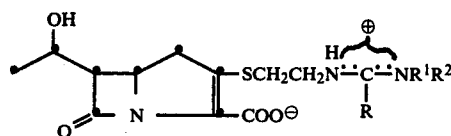
The substituted N-methylene thienamycin derivatives of the present invention may be depicted by the following structural formula:



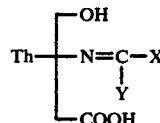
which, depending upon the basicity of the amino nitrogen (a function of the identity of the methylene substituents X and Y), may equivalently be represented as an inner salt:



which is one canonical form of a single resonant structure, which, for example, when Y is $\text{—NR}^1\text{R}^2$ and X is R is:



For convenience, the compounds of the present invention may be represented by the symbol:

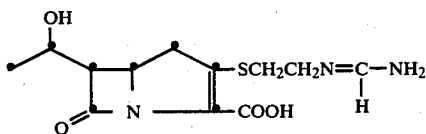


wherein "Th" indicates the bicyclic nucleus of thienamycin and its hydroxyl, amino, and carboxyl functional groups are shown; and wherein X and Y are independently selected from the group consisting of hydrogen, —R , —OR , —SR , and $\text{—NR}^1\text{R}^2$; R^1 and R^2 are independently selected from R, hydrogen, nitro, hydroxyl, alkoxy having 1–6 carbon atoms, amino, mono- di- and trialkylamino wherein the alkyl moieties each comprise 1–6 carbon atoms; R^1 and R^2 may be joined together to form a substituted or unsubstituted mono- or bicyclic heteroaryl or heterocyclyl comprising (together with the nitrogen atom to which they are attached) 4–10 atoms one or more of which may be an additional hereto atom selected from oxygen, sulphur or nitrogen; R, R^1 and R^2 are substituted or unsubstituted: cyano; carbamoyl; carboxyl; alkoxy carbonyl and alkyl having from 1 to about 10 carbon atoms; alkenyl having from 2 to about 10 carbon atoms; alkynyl having from 2 to about 10 carbon atoms; cycloalkyl having from 3 to 10 carbon atoms; cycloalkylalkyl and cycloalkylalkenyl having from 4 to 12 carbon atoms; cycloalkenyl, cycloalkenylalkenyl, and cycloalkenylalkyl having 3–10, 4–12 and 4–12 carbon atoms, respectively; aryl having from 6 to 10 carbon atoms, aralkyl, aralkenyl, and aralkynyl having from 7 to 16 carbon atoms; mono- and bicyclic heteroaryl and heteroaralkyl which typically comprise 4 to 10 ring atoms one or more of which is a hetero atom selected from oxygen, sulphur, or nitrogen and wherein the alkyl moiety of the heteroaralkyl radical comprises 1 to about 6 carbon atoms; mono- and bicyclic heterocyclyl and heterocyclylalkyl which typically comprises 4 to 10 ring atoms one or more of which is a hetero atom selected from oxygen, sulphur or nitrogen and wherein the alkyl moiety of the heterocyclylalkyl radical comprises from 1 to about 6 carbon atoms; and wherein the above-mentioned substituent or substituents on R, R^1 , R^2 or on the ring formed by the joinder of R^1 and R^2 , are selected from the group consisting of: halo, such as chloro, bromo, iodo and fluoro; azido; alkyl having 1–4 carbon atoms; thio; sulpho; phosphono; cyanothio (—SCN); nitro; cyano; amino; hydrazino; mono-, di- and trialkyl substituted amino, and hydrazino wherein the alkyl has 1–6 carbon atoms; hydroxyl; alkoxy having 1–6 carbon atoms; alkylthio having 1–6 carbon atoms; carboxyl; oxo; alkoxy carbonyl having 1–6 carbon atoms in the alkoxy moiety; acyloxy comprising 2–10 carbon atoms; carbamoyl and

and magnetically stirred at 22° C. After 1 hour the crude reaction is chromatographed on a 250 μ thick silica gel plate developed in 8:2, chloroform, ethanol. The band of R_f 0.1– R_f 0.3 is removed and eluted with ethanol. Benzimidoyl thienamycin, 3-methyl-2-buten-1-yl ester hydrobromide is isolated as a solid after precipitation from an ethanol-chloroform solution with hexane.

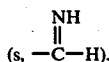
EXAMPLE 5

Preparation of N-Formimidoyl Thienamycin



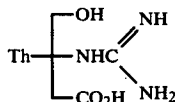
Thienamycin (517 mg) is dissolved in pH 7 0.1 N phosphate buffer (25 ml) and cooled in an ice bath with magnetic stirring. The solution is adjusted to pH 8.5 using 2.5 N sodium hydroxide solution dispensed from an automatic burette. While maintaining a pH of 8.5, methyl formimide hydrochloride (711 mg) is added portionwise over 2–3 minutes. After an additional 10 minutes, the pH of the solution is brought to 7.0 using 2.5 N hydrochloric acid. The solution is chromatographed on a column of XAD-2 resin (150 cc) which is eluted with water. The N-formimidoyl thienamycin derivative elutes in 1.5–2.0 column volumes (200–300 cc) and is lyophilized to a white solid (217 mg).

UV (pH 7 0.1 N phosphate buffer) λ_{max} 297 nm (8,590). ir (Nujol mull) 1767 cm^{-1} (β -lactam) nmr (D_2O) δ 1.37 (d, $J=6$ Hz, CH_3-CH), 3.0–3.75 (m, $-CH_2-$), 4.2–4.8 (m, C_5H , C_6H , C_7H), 7.86



Following the procedure of Example 5, enhanced product isolation is achieved when the XAD-2 column is replaced by an otherwise equivalent column of Dowex 50-X4 (Na^+ cycle, 200–400 mesh).

EXAMPLE 6

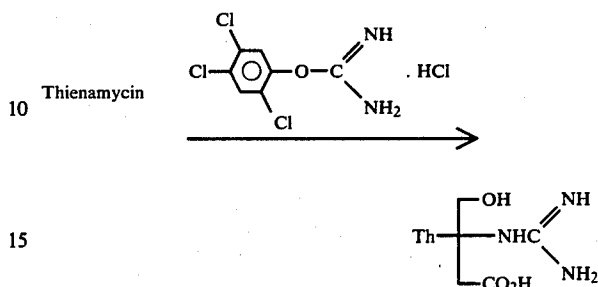


PREPARATION OF N-GUANYL THIENAMYCIN

Thienamycin (8.9 mg.) is dissolved in pH 7 0.1 N phosphate buffer (0.7 ml.) and N,N-dimethylformamide (0.3 ml.) and the solution brought to pH 9.5 by the addition of 2.5 N sodium hydroxide solution. To the magnetically stirred solution is added O-methylisourea hydrochloride (43 mg.) causing a slight drop in pH. Additional sodium hydroxide solution is added to bring the pH back to 9.5 and the solution is stirred 30 minutes at 23° C. The solution is then acidified to pH 7.0. A sample of the solution containing a mixture of thienamycin and N-guanyl thienamycin shows two bioactive zones after electrophoresis (50 V/cm., 20 minutes, 0.05 N pH 7

phosphate buffer) and bioautography on *S. aureus* plates.

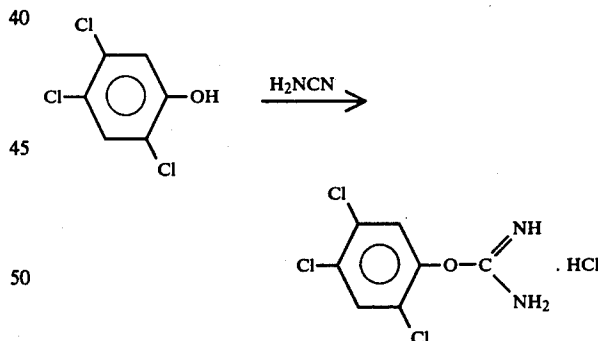
EXAMPLE 7



PREPARATION OF N-GUANYL THIENAMYCIN

Thienamycin (11 mg.) is dissolved in pH 7 0.1 N phosphate buffer (1 ml.) and adjusted to pH 8.3 with 0.1 N sodium hydroxide by means of an automatic dispensing burette. To the magnetically stirred solution is added 0-2,4,5-trichlorophenylisourea hydrochloride (76 mg.) portionwise to allow the auto burette to maintain a nearly constant pH. The reaction is run 4 hours at 22° C. and is then readjusted to pH 7.0 by the addition of dilute acid. A sample of this solution containing thienamycin and N-guanyl thienamycin is electrophoresed (50 V/cm., 25 minutes, pH 7 0.1 N phosphate buffer) and shows a positive Sakaguchi spray zone at 2.0 cm. towards the anode and a positive ninhydrin spray zone at 1.5 cm. in the same direction.

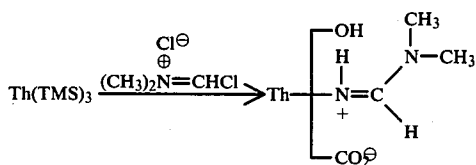
EXAMPLE 8



PREPARATION OF 0-2,4,5-TRICHLOROPHENYLISOUREA.HYDROCHLORIDE

A solution of cyanamide (0.28 mg.) in ether (0.50 ml.) is mixed with 2,4,5-trichlorophenol (12.5 g.); the mixture is heated to 70° C. and the melt magnetically stirred while the reaction flask is flushed with nitrogen. Dry halogen chloride gas is then slowly bubbled into the melt and the reaction is allowed to cool to 22° C. The resulting solid is washed thoroughly with ether and filtered to give 0-2,4,5-trichlorophenylisourea hydrochloride as a white solid, m.p. 205°–206° C.

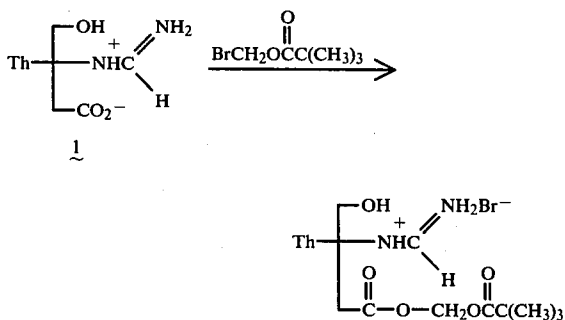
EXAMPLE 9



PREPARATION OF
N',N'-DIMETHYLAMINOMETHYLENE
THIENAMYCIN

Thienamycin (16.5 mg.) is silylated with hexamethyldisilazane (200 $\mu\text{l.}$) and trimethylchlorosilane (60 $\mu\text{l.}$) in the usual manner. The silylated thienamycin is suspended in (ethanol free) chloroform (1 ml.) with magnetic stirring under a nitrogen atmosphere. The mixture is cooled to -45°C. and a solution of triethylamine (21 $\mu\text{l.}$) in chloroform (21 $\mu\text{l.}$) is added followed by a solution of (chloromethylene)-dimethylammonium chloride (11.5 mg.) in chloroform (50 $\mu\text{l.}$). The mixture is warmed to -25°C. over 1 hour and 0.1 N pH 7 phosphate buffer (5 ml.) is added. The mixture is vigorously stirred 15 minutes. The aqueous phase is separated and contains N-dimethylaminomethylene thienamycin which has an electrophoretic mobility (50 V/cm., 1 hour, pH 7 buffer) of 3.6 cm. towards the cathode.

EXAMPLE 10

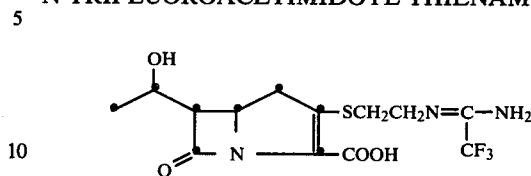


PREPARATION OF N-FORMIMIDOYL
THIENAMYCIN PIVALOXYMETHYL ESTER
HYDROBROMIDE

N-aminomethylene thienamycin **1** (10 mg.) is dissolved in hexamethylphosphoramide (200 $\mu\text{l.}$) containing bromomethyl pivalate (10 $\mu\text{l.}$) and triethylamine (1 $\mu\text{l.}$) and magnetically stirred at 22°C. After 2 hours the hexamethylphosphoramide solution is dissolved in 2 ml. methylene chloride and the product precipitated with a 50% hexane-ether solution. The precipitate is dissolved in an aqueous 10% tetrahydrofuran solution and chromatographed on an XAD-2 resin packed column. N-Formimidoyl thienamycin pivaloxymethyl ester is isolated as a solid after tetrahydrofuran elution of the column and lyophilization.

EXAMPLE 11

PREPARATION OF
N-TRIFLUOROACETIMIDOYL THIENAMYCIN

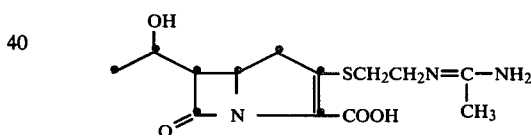


Thienamycin (199 mg) is dissolved in pH 7 0.1 N phosphate buffer (7 ml) and adjusted to pH 8.5 with 1 N sodium hydroxide solution. While maintaining this pH with an automatic burette, a solution of methyl trifluoroacetimidate (355 $\mu\text{l.}$) in dioxane (2.5 ml) is added at once. After 30 minutes the pH is readjusted to 7.0 by the addition of 1 N hydrochloric acid. The solution is then chromatographed on Dowex 50-X4 resin (200 cc, Na^+ cycle, 200-400 mesh) and is eluted with water. The N-trifluoroacetimidoyl Thienamycin derivative elutes in the first half column volume. This eluate is rechromatographed in a similar manner on Dowex 50-X4 (100 cc, Na^+ cycle, 200-400 mesh) and the first column volume concentrated and chromatographed on XAD-2 resin (30 cc). The N-trifluoroacetimidoyl Thienamycin derivative elutes in 2.5-5.0 column volumes which is lyophilized to a white solid (15 mg). UV (pH 7 0.1 N phosphate buffer) λ_{max} 302 nm (ϵ 4,450). ir (Nujol mull) 1750 cm^{-1} (β -lactam).

Electrophoresis: (50 v/cm, 20 min, pH 7, 0.1 N phosphate buffer) mobility 2.0 cm (toward cathode)

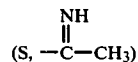
EXAMPLE 12

Preparation of N-Acetimidoyl Thienamycin



Thienamycin (190 mg) is dissolved in pH 7 0.1 N phosphate buffer (13 ml) and cooled in an ice bath with magnetic stirring. The solution is adjusted to pH 8.5 using 2.5 N sodium hydroxide solution dispensed from an automatic burette. While maintaining a pH of 8.5, ethyl acetimidate hydrochloride (400 mg) is added portionwise over a few minutes. After an additional 40 minutes the solution is adjusted to pH 7.0 with 2.5 N hydrochloric acid. The solution is then chromatographed on Dowex 50-X8 resin (250 cc, Na^+ cycle, 100-200 mesh) and is eluted with water. The N-acetimidoyl derivative elutes in 1-2 column volumes (240-520 cc) and is lyophilized to a white solid (88 mg). UV (pH 7 0.1 N phosphate buffer) λ_{max} 297 nm (ϵ 7,620). ir (Nujol mull) 1774 cm^{-1} , β -lactam.

nmr (D_2O) δ 1.27 (d, $J=6\text{ Hz}$, CH_3-CH) 2.24



3.2-3.5 (m, $-\text{CH}_2-$), 3.5-3.9 (m, $-\text{CH}_2-$) 4.2-4.6 (m; C_5H , C_6H , C_7H).